Multi Client Chatting - TCP and UDP Communication in ns-3

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Lab Task:

Implement a multi-client chatting system using both **TCP and UDP protocols**. Measure and analyze the **network performance** during communication. The **client should send input arguments to the server**, the **server will perform the required calculations**, and the **results should be returned and displayed on the client side**. Refer to the **model code and sample output provided in Appendix 1**.

OUTPUT:

```
--- Simulation Start ---
Server received message from 10.1.1.1
Server received message from 10.2.1.1
[CLIENT_A] Received -> ACK | Received at 0.0020784 s | Size 19 bytes
[CLIENT_B] Received -> ACK | Received at 0.0020784 s | Size 19 bytes
Server received message from 10.1.1.1
[CLIENT_A] Received -> ACK | Received at 3.00208 s | Size 19 bytes
Server received message from 10.2.1.1
[CLIENT_B] Received -> ACK | Received at 5.00208 s | Size 19 bytes
Server received message from 10.1.1.1
[CLIENT_A] Received -> ACK | Received at 6.00208 s | Size 19 bytes
Server received message from 10.1.1.1
[CLIENT_A] Received -> ACK | Received at 9.00208 s | Size 19 bytes
Server received message from 10.2.1.1
[CLIENT_B] Received -> ACK | Received at 10.0021 s | Size 19 bytes
Server received message from 10.1.1.1
[CLIENT_A] Received -> ACK | Received at 12.0021 s | Size 19 bytes
--- Simulation End ---
```